

UNIVERSITY OF MAURITIUS

FACULTY OF INFORMATION, COMMUNICATION AND DIGITAL TECHNOLOGIES



SECOND SEMESTER EXAMINATIONS

JULY/AUGUST 2021

PROGRAMME	BSc (Hons) Information Systems		
MODULE NAME	Information Systems Security		
DATE	Monday 09 August 2021	MODULE CODE	SIS3118Y
TIME	13.30-15.30 hrs	DURATION	2 Hours
NO. OF QUESTIONS SET	6	NO. OF QUESTIONS TO BE ATTEMPTED	4

INSTRUCTIONS TO CANDIDATES

This paper contains 2 sections A and B.

Answer any 2 questions from each section.

All questions carry equal marks.

SECTION A

Answer any 2 questions.

Use separate answer book for each section

Question 1

- (a) Name five main concepts aim at in a secure system.
- (b) Using suitable examples of your own explain the concepts mentioned in (i) and their respective protection mechanism.
- (c) As a staff responsible for creating a secure environment; what are the steps you would take to ensure an adequate security level in an organization?
- (d) Comment on the need for a security policy.
- (e) Classical ciphers were weak to frequency analysis. Comments on the operations included in modern symmetric ciphers to cater for the deficiencies. Use suitable examples of your own.

[5*1 + 3 + 6* 1 + 4 + 7 marks]

(Total: 25 Marks)

Question 2

- (a) A bank ATM system relies on a secret code of 4 digits. Calculate the work factor of this implementation? Explain all your steps
- (b) Differentiate between symmetric and public key cryptosystems?
- (c) State the evidences which confirm the high degree of diffusion and confusion with DES.
- (d) Using a labeled diagram explain operation of a hybrid system which also caters for non-repudiation and integrity?

(4 + 7 + 4 + 10 marks)

(Total: 25 Marks)

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Question 3

(a) Briefly explain the terms following terms

- (i) Malware
- (ii) Virus
- (iii) Hacking
- (iv) Scareware
- (v) Cracking

(b) What do you understand by the term network firewall?

(c) Name four devices that can perform firewall function.

(d) Most companies adopt the DMZ architecture given its flexibility and robustness against attack; draw a labelled diagram to explain its operation.

(e) Comment on the statement-“ Although we have invested massively in latest firewall technologies we are still getting security issues”-

(5* 1 + 3 + 4 + 8 + 5 marks)

(Total: 25 Marks)

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SECTION B

Answer any 2 questions.

Use separate answer book for each section

Question 4

- (a) Secret sharing is considered to be the de-facto standard for a number of applications. Comments on **SIX** basic precautions that you would recommend when creating the authentication code.

[6 marks]

- (b) Explain the term Intrusion and list **FOUR** ways used to intrude systems.

[3 + 4 marks]

- (c) There are three models of intrusion detection mechanisms: anomaly-based detection, signature-based detection, and hybrid detection. Distinguish between the anomaly-based detection, signature-based detection.

[6 marks]

- (d) Security breaches can affect an organization in different ways. Discuss **THREE** effects of security breaches.

[6 marks]

(Total: 25 Marks)

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Question 5

- (a) Wireless fidelity (wi-fi) networks are being deployed at a tremendous rate. Comment on the measures, other than password protection, you would implement to reinforce its security.

[4 marks]

- (b) List three different encryption mechanisms commonly used for Wi-fi networks.

[3 marks]

- (c) A customer makes the following comments to you “though we have implemented the most expensive firewall we are still having security problems”. As a security advisor, give plausible explanations and provide solutions.

[6 marks]

- (d) Identity of a person/entity can be protected using several techniques such as

- (i) Anonymity
- (ii) Pseudonymity
- (iii) Unobservability
- (iv) Unlinkability.

Explain how each of the above techniques are used.

[4x1 marks]

- (e) According to many professionals, Computer Forensics is a four (4) step process consisting of:

- (i) Acquisition
- (ii) Identification
- (iii) Evaluation
- (iv) Presentation.

Explain each of the above steps.

[4x2 marks]

(Total: 25 Marks)

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Question 6

- (a) What do you understand by the term watermark? Using an analogy to explain how digital watermark resembles real world watermark.

[4 marks]

- (b) Differentiate between steganography and watermarking?

[4 marks]

- (c) Give **four** uses of watermark.

[4 marks]

- (d) Name **three** biometric that are commonly used.

[3 marks]

- (e) List **five** important properties which an element should possess before its use in biometric authentication.

[5 marks]

- (f) Explain the operation of a biometric system in identification mode. Use block diagrams to simplify the steps.

[5 marks]

(Total: 25 Marks)

END OF QUESTION PAPER